MERON ZERIHUN DEMISSIE

Ph.D. Student | Computer Science and Engineering

mdemissi@umich.edu | meronzerihun.github.io | Google Scholar

EDUCATION

Ph.D. | Computer Science and Engineering | University of Michigan (UM) | January 2024 – Present (Expected date: Dec 2028)

Advisor: Ronald Dreslinski Jr. Area: Computer Architecture – Hardware Reliability, Privacy and Security

Masters | Computer Science and Engineering | University of Michigan (UM) | January 2022 – December 2023

B.Sc. | Software Engineering | Addis Ababa Institute of Technology (AAiT) | September 2016 – October 2021

EMPLOYMENT HISTORY

Graduate Student Instructor, University of Michigan, January 2024 – April 2024, August 2024 – Present

- EECS 370: Introduction to Computer Organization

Graduate Technical Intern, Intel Corporation, May 2024 – August 2024

- Worked on HEIR, an MLIR compiler that transforms classical applications to use homomorphic encryption.
- Mentors: Alexander Viand and Duhyeong Kim, Manager: Rosario Cammarota.

Graduate Student Staff Assistant, University of Michigan, January 2024 – April 2024

- CSE's DEI committee graduate student liaison

Graduate Student Instructor, University of Michigan, August 2023 – December 2023

- EECS 573: Advanced Topics in Computer Architecture

Graduate Student Research Assistant, University of Michigan, May 2022 – July 2023

PUBLICATIONS

Under Review 2026 | Dash: A Hardware Approach for Safe Data Disclosures | M. Zerihun Demissie, L. Biernacki, D. Ramotsoela, T. Austin. Proposes a hardware technique that allows authorized data disclosures for cloud computation under data privacy guarantees.

ArXiv 2025 | HEIR: A Universal Compiler for Homomorphic Encryption | A. Ali, J. Choi, B. Gipson, S. Gorantala, J. Kun, W. Legiest, L. Lim, A. Viand, **M. Zerihun Demissie**, H. Zheng. Presents a unified MLIR compiler framework that provides foundational homomorphic encryption (HE) abstractions to ease HE research and programming.

ISPASS 2023 | Exploring the Efficiency of Data-Oblivious Programs | L. Biernacki, B. Mengist Tiruye, **M. Zerihun Demissie**, F. Assamnew Andargie, B. Reagen, T. Austin. Explores the costs of data-oblivious transformations and the role of algorithm redesign in alleviating cost.

SEED 2022 | Sequestered Encryption: A Hardware Technique for Comprehensive Data Privacy | L. Biernacki, **M. Zerihun Demissie**, K. Birkayehu Workneh, F. Assamnew Andargie, T. Austin. Explores how trusted hardware can provide data confidentiality that is resilient under vulnerable software within practical performance overheads.

SEED 2021 | VIP-Bench: A Benchmark Suite for Evaluating Privacy-Enhanced Computation Frameworks | L. Biernacki, **M. Zerihun Demissie**, K. Birkayehu Workneh, G. Basha Namomsa, P. Gebremedhin, F. Assamnew Andargie, B. Reagen, T. Austin. Presents a benchmark suite to evaluate competing privacy frameworks fairly and directly.

SERVICE

CSEG Service Co-chair, August 2025 – May 2026

- Board member in the CSEG student organization, worked on inclusivity and voicing students' concerns.

Graduate Student Mentor | University of Michigan African Presidential Scholars UMAPS, January 2022 – May 2023

- Mentored for a project that enhances privacy computation with integrity.

Graduate Student Mentor | African Undergraduate Research Adventures AURA, May 2022 – July 2022

- Mentored for a project on interfacing the PALISADE homomorphic encryption library with VIP-Bench privacy-focused benchmark suite

---

HONORS AND AWARDS

- CSE DEI Service Awards, UM, 2024
- Certificate of Great Achievement, AAiT, 2021
- Award of Appreciation of Top 5 Female Students, AAiT, 2017

OUTREACH ACTIVITIES

Primary Instructor | Discover Engineering, University of Michigan, June 2022

- Coordinated a computer basics workshop for 8th–10th graders.

Student Mentor | Technovation Challenge w/ Ministry of Innovation and Technology, March 2020 – April 2020

- Mentored female high school students to develop software to solve outstanding problems within our community.

SKILLS

Programming Languages: C/C++, Python, Java, JavaScript, C#, Go, Kotlin, **Architectural Simulations and HDLs:** PIN, gem5, Verilog, **Compiler Extensions:** Clang, MLIR, LLVM, **Other Skills:** Bash Scripting, Database (SQL, MongoDB, Firebase), DBMS (MySQL, Oracle), Libraries (jQuery, NodeJS), Frameworks (Spring, ASP.Net, Ruby On Rails), Platforms (Android Studio, Visual Studio)

Research Statement (Cover Letter)

My research focuses on enhancing computer architecture to build trustworthy, reliable, and resilient computing systems. As modern computing scales, traditional hardware primitives that treat all workloads uniformly impose unacceptable performance overheads and fail to deliver robust dependability guarantees. My work addresses this challenge through workload-informed hardware co-design, leveraging application semantics, execution invariants, and numerical properties to design specialized hardware architectures that deliver strong security, integrity, and fault tolerance.

My earlier research applied workload-informed design to solve privacy and integrity challenges in secure cloud computing. To protect sensitive data from unauthorized disclosure, Sequestered Encryption (SE) uses hardware to enforce a data-oblivious execution model to eliminate microarchitectural side-channel leakage. Extending this foundation, my follow-up work, DASH, shifted remote cloud computing toward verifiable, proof-based execution. DASH uses hardware to generate execution evidence, allowing external users to validate result integrity and selectively disclose specific computation outputs without compromising underlying data confidentiality. Together, SE and DASH demonstrate how hardware mechanisms tailored to program execution rules can guarantee robust privacy and integrity.

My dissertation research applies this same core philosophy to solve reliability challenges in large-scale AI infrastructure. High Bandwidth Memory (HBM) failures in GPUs are a primary driver of costly LLM training interruptions, yet existing hardware reliability schemes treat memory uniformly and operate agnostically to deep learning numerics. Through fault-injection experiments across mixed-precision training formats (FP16, BF16, FP8) and optimizer states, we discovered that fault vulnerability in stored training state is fundamentally determined by value rather than bit position alone. Because parameter values follow structured distributions, critical vulnerability is highly sparse and dynamically tracks numerical magnitude. Leveraging this finding, my proposed research explores state-aware memory protection architectures for GPUs that match hardware protection to the numerical resilience of stored parameters, preventing silent data corruption and converting job-killing stops into soft, software-assisted repairs.

Looking forward, I am passionate about computer architecture research at the intersection of system reliability, and hardware security. As AI datacenters scale across thousands of GPUs, workload-agnostic hardware primitives will become increasingly unsustainable. My goal is to continue developing intelligent hardware primitives that adapt to application dynamics, bridging deep learning behavior with specialized hardware to enable continuous, reliable, and trustworthy computing at scale.